

Md. Raihan Siddik

Computer Vision | Signal & Image Analysis | Machine Learning | Multimodal Learning | Biomedical AI

✉ raihansiddik2006120@gmail.com ☎ +880 1580 873401 📍 Dhaka, Bangladesh

🌐 LinkedIn 🐙 GitHub 📁 Portfolio

RESEARCH PROFILE

Electrical and Electronic Engineering graduate from BUET, specializing in Communication and Signal Processing, with research experience in computer vision, parameter-efficient learning, and efficient deep-learning models. Worked on frequency-aware Vision Transformer adaptation, CLIP-LoRA few-shot anomaly detection, wavelet-based image super-resolution, semantic segmentation, and audio-visual learning. Experienced in LoRA, Vision Transformers, CLIP, frequency-domain analysis, cross-dataset generalization, and data-efficient model adaptation. Interested in extending this work toward vision-language models (VLMs), multimodal large language models (MLLMs), multimodal learning, and trustworthy AI. Also interested in applying computer vision and multimodal learning to medical imaging, biomedical signal and image analysis, and healthcare-related AI problems.

RESEARCH INTERESTS

Computer Vision; Parameter-Efficient Fine-Tuning (PEFT); Efficient AI; Vision Transformers; Vision-Language Models (VLMs); Multimodal Learning; Multimodal Large Language Models (MLLMs); Trustworthy AI; Robust and Generalizable Machine Learning; Medical Image Analysis; Biomedical Signal & Image Analysis; Machine Learning for Healthcare.

EDUCATION

Bangladesh University of Engineering and Technology (BUET), Dhaka	June 2026
<i>B.Sc. in Electrical and Electronic Engineering; Major: Communication and Signal Processing</i>	<i>CGPA: 3.54/4.00</i>

- Last two semesters: 3.84/4.00.
- **Relevant coursework:** Artificial Intelligence and Machine Learning; Artificial Intelligence and Machine Learning Laboratory; Introduction to Medical Imaging; Probability and Statistics; Linear Algebra; Numerical Techniques Laboratory; Computer Programming and Sessional; Continuous Signals and Linear Systems; Digital Signal Processing I and Laboratory; Random Signals and Processes.

Higher Secondary Certificate (HSC), Dinajpur Board	2020
GPA: 5.00/5.00	

Secondary School Certificate (SSC), Dinajpur Board	2018
GPA: 5.00/5.00	

PUBLICATIONS & MANUSCRIPTS

- **FaLoRA: Parameter-Efficient Frequency-Aware Adaptation for Generalizable Deepfake Detection.** First-author manuscript; submitted to IEEE Transactions on Dependable and Secure Computing (TDSC); under review.
- **Image Super-Resolution Using Wavelet Residual Convolutional Neural Networks.** Second-author manuscript; submitted to ICECE 2026; under review.
- **CLIP-LoRA for Few-Shot Industrial Anomaly Detection.** First-author; submitted to ICECE 2026; under review.

RESEARCH EXPERIENCE

FaLoRA: Frequency-Aware Parameter-Efficient Deepfake Detection	Undergraduate Thesis
<i>Supervisor: Dr. S. M. Mahbubur Rahman, Professor, Department of EEE, BUET</i>	

- Developed a frequency-aware LoRA framework combining DCT, FFT phase, and steganalytic noise features to adapt a Vision Transformer while updating approximately 1% of the model parameters.
- Designed preprocessing, training, and evaluation pipelines and achieved cross-dataset AUC scores of 90.96% on Celeb-DF v2, 79.34% on DFDC, 80.95% on WildDeepfake, and 89.26% on DeeperForensics.
- Conducted generalization analysis across unseen datasets, compared parameter efficiency, performed ablation studies, and submitted the first-author manuscript to IEEE TDSC; under review.

Image Super-Resolution Using Wavelet Residual Convolutional Neural Networks	Second-Author Manuscript
<i>WaveSRResNet: spatial-wavelet learning for high-fidelity image reconstruction</i>	

- Contributed to a Wavelet Residual Block that combines spatial-domain convolution with forward and inverse discrete wavelet transforms to preserve edges and high-frequency textures.
- Evaluated the model on standard super-resolution benchmarks, achieving up to 4.8% higher PSNR and 8% higher SSIM for 8× scaling, with a 9% LPIPS reduction for 4× scaling.
- Analyzed spectral bias, edge sharpness, perceptual fidelity, and reconstruction quality; the manuscript was submitted to ICECE 2026; under review.

- Developed a parameter-efficient approach for adapting the CLIP visual encoder with LoRA using a limited number of labeled normal and defective images.
- Built a generalizable image-text evaluation workflow focused on representation quality, computational efficiency, and transfer to unseen defect patterns and product categories.
- Conducted comparative experiments and ablation studies; the first-author manuscript is under review at ICECE 2026.

SELECTED RESEARCH & ENGINEERING PROJECTS

Python, PyTorch, TensorFlow Lite

- Built a spatial audio-separation pipeline using DP-PCRN and Causal CRN with interaural phase and level difference features; achieved PESQ 3.05 and STOI 0.906.
- Optimized inference for mobile deployment using TensorFlow Lite int8 quantization, connecting model development with practical efficiency constraints.

- Developed an end-to-end visual pipeline for aerial image acquisition, disease classification, semantic segmentation, and vegetation-index heatmap generation.
- Achieved 97.7% pixel accuracy with a SegFormer-based model and integrated the workflow with embedded hardware and autonomous flight control.

TECHNICAL SKILLS

Programming	Python, MATLAB, C, C++
Deep Learning	PyTorch, TensorFlow, Keras, CNNs, Vision Transformers, CLIP, LoRA, transfer learning, model quantization
Computer Vision / Imaging	Image classification, semantic segmentation, anomaly detection, deepfake detection, image super-resolution, image preprocessing, medical-imaging foundations, OpenCV
Scientific Computing	NumPy, pandas, scikit-learn, signal processing, frequency-domain analysis, feature extraction
Research Methods	Dataset curation, cross-dataset validation, ablation studies, error analysis, reproducible experimentation, scientific writing
Evaluation	ROC/AUC, confusion-matrix analysis, imbalanced classification, PSNR, SSIM, LPIPS, PESQ, STOI
Tools / Platforms	GitHub, Linux, Google Colab, LaTeX, TensorFlow Lite, Raspberry Pi, ArduPilot

INDUSTRIAL EXPOSURE

- Gained exposure to network monitoring, IP transit and peering, operational infrastructure, and reliability-focused technical workflows.

REFERENCES

Dr. S. M. Mahbubur Rahman Professor, Dept. of EEE, BUET ✉ mahbubur@eee.buet.ac.bd ☎ +880 1724 588 513	Dr. Md. Kawsar Alam Professor, Dept. of EEE, BUET ✉ kawsaralam@eee.buet.ac.bd ☎ +880 1712 526 059
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